

Dot products and weighted sums

Project: Image-pattern Network Investigation

1. Prepare the purpose, prerequisites and materials

Teach the learning cycle

Primary target: Calculate a dot product by matching feature positions

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Representing inputs as vectors

Without opening the old lesson, explain one key idea from Representing inputs as vectors. Give an example and a mistake that the idea helps you avoid.

Look for: Encode a record using a fixed feature order and units

Reference example, not the only acceptable wording: Teacher explanation: A [4, 2], B [2, 4], C [3, 3]. Each has dimension 2. The table has 3 rows and 2 columns.

Teach the dot product as a traceable sum of signed contributions in two 45-minute sessions. Keep weights fixed throughout each comparison and separate hand-chosen parameters from training. Students should explain a score change using the corresponding coefficient, while retaining the limits of that interpretation.

Outcomes

Calculate a dot product by matching feature positions

Explain how a weight changes a score with other inputs fixed

Distinguish a weighted score from a probability or trained model

Preparation

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Prerequisite answer

Their components assign different values to width and height. A calculation using different weights for those features can produce different results.

End goal for the pictorial case

By the end, I can explain dot products and weighted sums using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain and model the mechanism

A small model can combine several measurements into one score. Let $x = [4, 2]$ contain fictional width and height measurements in cm. Set weights $w = [2, -1]$. Multiply matching positions, then add: $s = (2 \text{ times } 4) + (-1 \text{ times } 2) = 6$. The letter x names the input vector.

We call this calculation a dot product, written $w \cdot x$. The vectors must have the same number of components. Each weight belongs to one named feature. These hand-selected weights demonstrate a computation; no training has happened. The result 6 is a score, not a probability or a claim that this parcel is safe.

The coefficients carry score-per-cm units in this example, making both contributions compatible score units. A weighted sum needs a consistent definition of units.

Keep $w = [2, -1]$ fixed. For $A = [4, 2]$, the score is 6. For $B = [2, 4]$, it is $4 - 4 = 0$. For $C = [3, 3]$, it is $6 - 3 = 3$. Swapping input positions without swapping their matching weights changes the computation. A zero score can come from cancellation, even when neither input is zero.

The length check matters: Python zip alone stops at the shorter list and could silently omit a feature. The function assumes numeric inputs already validated under the schema. First predict the three printed values on paper, then compare.

Term · Meaning

Term: Weight

Meaning: A numerical coefficient multiplying one input component

Term: Dot product

Meaning: The sum of products of corresponding vector components

Term: Contribution

Meaning: The product of one input component and its matching weight

Term: Score

Meaning: A numerical output whose interpretation must be specified

Multiply matching components then add

Position · Contribution rule · Worked contribution

Position: 1 width

Contribution rule: 2 times width

Worked contribution: 2 times 4 gives 8

Position: 2 height

Contribution rule: -1 times height

Worked contribution: -1 times 2 gives -2

Position: Add both

Contribution rule: 8 plus -2

Worked contribution: Score 6

The first two rows are separate contributions. The last row combines them; multiplying the final input sum by the final weight sum would be a different operation.

Guided: [4, 3] gives 8 and -3, total 5, down 1 from 6 because height rose by 1 with coefficient -1. Exit: the score has no supplied probability mapping, fitted probability model or calibration evidence. Multiplying it by ten does not create a supported probability.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Multiply matching components then add

Position · Contribution rule · Worked contribution

Position: 1 width

Contribution rule: 2 times width

Worked contribution: 2 times 4 gives 8

Position: 2 height

Contribution rule: -1 times height

Worked contribution: -1 times 2 gives -2

Position: Add both

Contribution rule: 8 plus -2

Worked contribution: Score 6

The first two rows are separate contributions. The last row combines them; multiplying the final input sum by the final weight sum would be a different operation.

Keep $w = [2, -1]$. Change $[4, 2]$ to $[4, 3]$. Compute both contributions and explain the score change.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Dot products and weighted sums

Draw each input's contribution to a toy weighted brightness score.

The facts we will use

Inputs $[2, 5]$; weights $[3, -1]$; no bias. Score is $2 \times 3 + 5 \times (-1)$.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Weighted brightness score

Input 2 x weight 3 = 6

Input 5 x weight -1 = -5

Add the products and bias 0: total = 1

Keep the total (identity) -> output 1

Dot products and weighted sums

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Multiply first input

Left contribution= $2 \times 3=6$.



2. Multiply second input

Right contribution= $5 \times (-1)=-5$.



3. Add contributions

Weighted sum= $6-5=1$; this is a score, not a probability.

CHANGE ONE THING

Increase only the right input from 5 to 6.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Multiply first input

What happens and why: Left contribution= $2 \times 3=6$.

Step: 2. Multiply second input

What happens and why: Right contribution= $5 \times (-1)=-5$.

Step: 3. Add contributions

What happens and why: Weighted sum= $6-5=1$; this is a score, not a probability.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A weight controls how strongly an input contributes to the score. The left connection adds three times its measurement; the negative right connection subtracts its measurement. Calculate each contribution before adding them. This dot product is a signed score; it becomes a probability only if a separate, justified transformation and interpretation are supplied.

What this example does not establish

These declared weights are hand-chosen toy parameters; no network training or calibrated probability is claimed.

Change one thing in the pictured case

Change: Increase only the right input from 5 to 6.

Prediction: The score becomes $6-6=0$.

Reason: The right weight is -1, so adding 1 to that input lowers the score by 1.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Time or stage · Teacher action

Time or stage: Session 1 start to 7 minutes

Teacher action: Ask the prerequisite question. Check signed multiplication with -1 times 2 before introducing a dot product.

Time or stage: 7 to 18 minutes

Teacher action: Lay out the contribution table. Students draw a connection between each named input and its own weight.

Time or stage: 18 to 32 minutes

Teacher action: Complete W1 and W2. Trace A, B and C with the same weights; discuss cancellation rather than calling zero an empty input.

Time or stage: 32 to 45 minutes

Teacher action: Run or trace the guarded Python function. Try a missing weight and explain the error. Complete the guided change.

Time or stage: Session 2 start to 8 minutes

Teacher action: Retrieve the two contributions for A. Ask what changes if height increases while width stays fixed.

Time or stage: 8 to 20 minutes

Teacher action: Complete W3, converting only height to mm. Show why weight magnitude depends on units.

Time or stage: 20 to 35 minutes

Teacher action: Collect W4 independently. Require the three signed contributions, original and changed totals, and the dimension error explanation.

Time or stage: 35 to 45 minutes

Teacher action: Complete W5 and the exit response. Use the fresh retry after targeted feedback; keep first attempts for assessment.

Use the worked example in Sections 3 and 6. Collect a prediction before revealing its result; ask the learner to point to each intermediate fact.

Provide three labelled contribution boxes and a number line for signed addition. Extension: permute both inputs and matching weights together and show that the sum is preserved; then permute only one to expose a schema mismatch.

Reduce help deliberately

Completion task: For $w = [2, -1]$ and $x = [2, 4]$, explain why score zero does not mean the input contains no information.

Teacher check: Teacher explanation: Contributions 4 and -4 cancel. The input still contains width 2 and height 4; zero is the combined score.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Dot products and weighted sums. Proposed explanation: “Multiply the sums instead”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The operation is 2 times 4 plus -1 times 2, not $(2 - 1)$ times $(4 + 2)$. That shortcut happens to match for some inputs but is not the rule.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Which connection explains why more right brightness lowers this score? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

The right connection has negative weight-1. Its contribution changes from-5 to-6 while the left contribution stays6. Changed-case check: The score becomes $6-6=0$. The right weight is-1, so adding1 to that input lowers the score by1.

Changed-case reasoning: The score becomes $6-6=0$. The right weight is-1, so adding1 to that input lowers the score by1.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the investigation and diagnose misconceptions

Multiply the sums instead: The operation is 2 times 4 plus -1 times 2, not $(2 - 1)$ times $(4 + 2)$. That shortcut happens to match for some inputs but is not the rule.

Negative means unimportant: A negative coefficient changes direction. Units, data and the rest of the model affect interpretation.

The weights have been learned: These values were supplied. Training would require data, an objective and a fitting procedure.

Use Section 9 for the evidence ledger. Add a boundary or missing-input case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Guided and exit questions

Guided: [4, 3] gives 8 and -3, total 5, down 1 from 6 because height rose by 1 with coefficient -1. Exit: the score has no supplied probability mapping, fitted probability model or calibration evidence. Multiplying it by ten does not create a supported probability.

W1 Calculate contributions

For $w = [2, -1]$ and $x = [3, 5]$, write both contributions and their sum.

Teacher explanation: Contributions are 6 and -5, so the score is 1.

W2 Explain cancellation

For $w = [2, -1]$ and $x = [2, 4]$, explain why score zero does not mean the input contains no information.

Teacher explanation: Contributions 4 and -4 cancel. The input still contains width 2 and height 4; zero is the combined score.

W3 Preserve a unit change

Width stays in cm and height changes to mm. For the worked parcel [4 cm, 20 mm], give the revised weights and score.

Teacher explanation: Use $[2, -0.1]$. Contributions 8 and -2 give score 6, preserving the original measurement meaning.

W4 Compute a new dot product independently

Use $x = [3, 1, 2]$ and $w = [-2, 4, 0.5]$. Give each contribution and the total. Increase only the second component from 1 to 2 and compute the new total. Explain why a two-weight list is invalid for this input.

Teacher explanation: Contributions -6, 4, 1 give -1. With second component 2, contributions -6, 8, 1 give 3, a rise of 4. Three input components need three matching weights; silently dropping the final component changes the computation.

W5 Find lost information

Verify that $[4, 2]$ and $[5, 4]$ both score 6 with $w = [2, -1]$. Can a later step receiving only this score distinguish those two records?

Teacher explanation: The scores are $8 - 2 = 6$ and $10 - 4 = 6$. A deterministic later step receiving only 6 receives identical information and cannot distinguish them.

Independent assessment and fresh retry

W4 requires contributions -6, 4, 1 and total -1, changed total 3 and a dimension check. Fresh retry: $w = [1, -3, 2]$, $x = [2, 1, 4]$ gives $2 - 3 + 8 = 7$. If only the third input falls to 3, the score is 5. Ask for the two-point decrease explanation.

Assess independent transfer

Selected case: Use $x = [3, 1, 2]$ and $w = [-2, 4, 0.5]$. Give each contribution and the total. Increase only the second component from 1 to 2 and compute the new total. Explain why a two-weight list is invalid for this input.

Selected case answer

Teacher explanation: Contributions -6, 4, 1 give -1. With second component 2, contributions -6, 8, 1 give 3, a rise of 4. Three input components need three matching weights; silently dropping the final component changes the computation.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect the project, support and next lesson

Add a labelled dot products and weighted sums evidence sheet to Image-pattern Network Investigation. Show how you can calculate a dot product by matching feature positions. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide three labelled contribution boxes and a number line for signed addition. Extension: permute both inputs and matching weights together and show that the sum is preserved; then permute only one to expose a schema mismatch.

Week 3 adds a bias after the dot product and uses an explicit decision boundary.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-5/3, ai-5/5, ai-5/8. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.